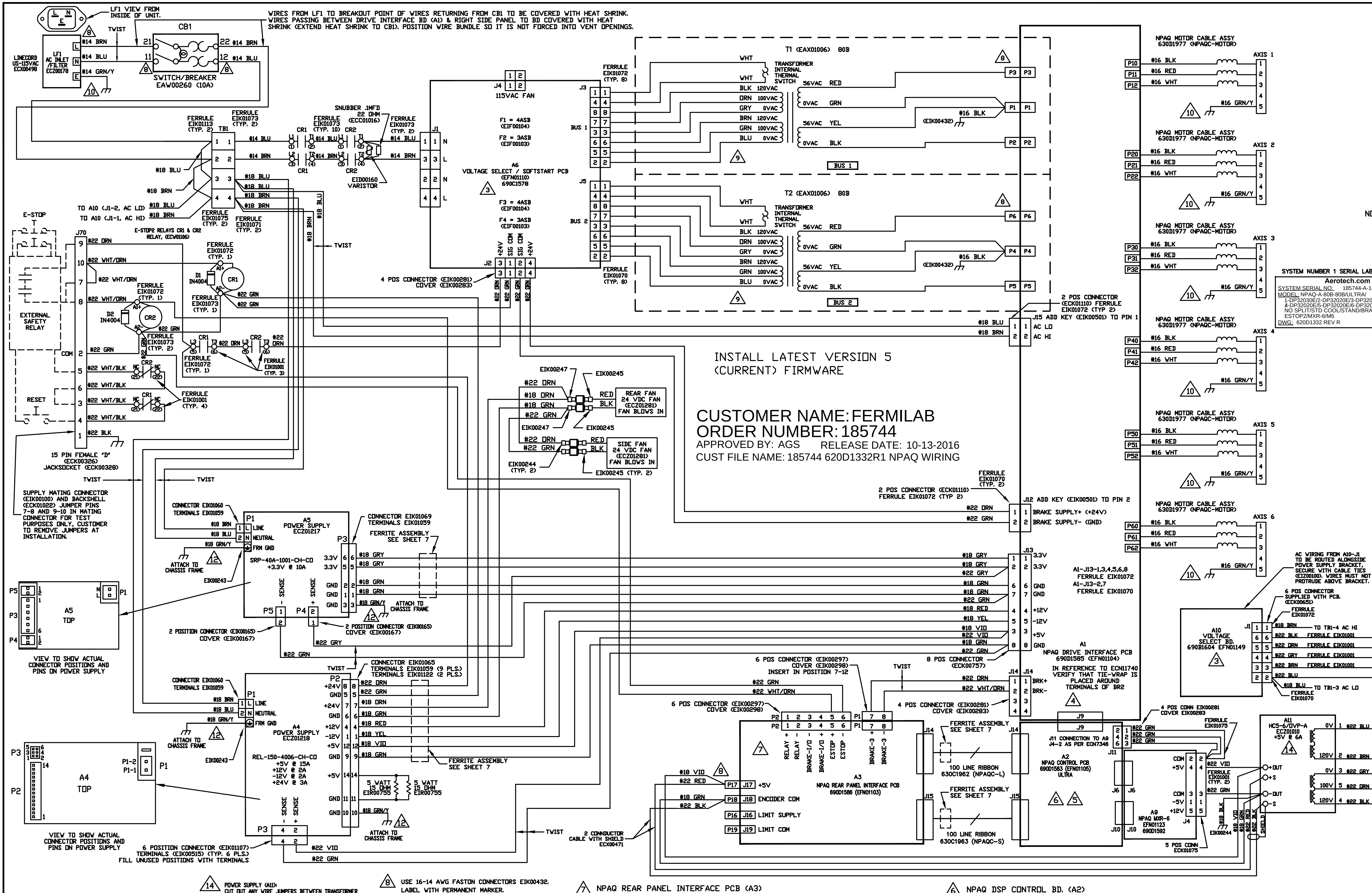
LF1 VIEW FROM INSIDE OF UNIT.
WIRES FROM LF1 TO BREAKOUT POINT OF WIRES RETURNING FROM CB1 TO BE COVERED WITH HEAT SHRINK. WIRES PASSING BETWEEN DRIVE INTERFACE BD (A1) & RIGHT SIDE PANEL TO BD COVERED WITH HEAT SHRINK (EXTEND HEAT SHRINK TO CB1). POSITION WIRE BUNDLE SO IT IS NOT FORCED INTO VENT OPENINGS.
LINECORD US-115VAC ECX00498
LF1 AC INLET /FILTER ECZ00178
CB1
SWITCH/BREAKER EAW00260 (10A)
TWIST
#14 BRN
#14 BLU
#14 GRN/Y
TB1
FERRULE EIK01113 (TYP. 2)
FERRULE EIK01073 (TYP. 2)
SNUBBER .1MFD 22 OHM (ECC01016)
EID00160 VARISTOR
J4 115VAC FAN
J1
A6 VOLTAGE SELECT / SOFTSTART PCB (EFN01110) 690C1578
F1 = 4ASB (EIF00104)
F2 = 3ASB (EIF00103)
F3 = 4ASB (EIF00104)
F4 = 3ASB (EIF00103)
J3
J5
BUS 1
BUS 2
T1 (EAX01006) 80B
T2 (EAX01006) 80B
TRANSFORMER INTERNAL THERMAL SWITCH
56VAC RED
0VAC GRN
56VAC YEL
0VAC BLK
120VAC
100VAC
0VAC
J2
4 POS CONNECTOR (EIK00281) COVER (EIK00283)
E-STOP
J70
EXTERNAL SAFETY RELAY
RESET
E-STOP2 RELAYS CR1 & CR2 RELAY, (ECV00106)
TO A10 (J1-2, AC LO)
TO A10 (J1-1, AC HI)
15 PIN FEMALE 'D' (ECK00326) JACKSOCKET (ECK00328)
SUPPLY MATING CONNECTOR (EIK00100) AND BACKSHELL (ECK01022) JUMPER PINS 7-8 AND 9-10 IN MATING CONNECTOR FOR TEST PURPOSES ONLY, CUSTOMER TO REMOVE JUMPERS AT INSTALLATION.
REAR FAN 24 VDC FAN (ECZ01281) FAN BLOWS IN
SIDE FAN 24 VDC FAN (ECZ01281) FAN BLOWS IN
A5 POWER SUPPLY ECZ01217
SRP-40A-1001-CH-CO +3.3V @ 10A
A5 TOP
VIEW TO SHOW ACTUAL CONNECTOR POSITIONS AND PINS ON POWER SUPPLY
A4 POWER SUPPLY ECZ01218
REL-150-4006-CH-CO +5V @ 15A +12V @ 2A -12V @ 2A +24V @ 3A
A4 TOP
5 WATT 15 OHM EIR00755
FERRITE ASSEMBLY SEE SHEET 7
ATTACH TO CHASSIS FRAME
6 POSITION CONNECTOR (EIK01107) TERMINALS (EIK00515) (TYP. 6 PLS.) FILL UNUSED POSITIONS WITH TERMINALS
2 CONDUCTOR CABLE WITH SHIELD ECX00471
A3 NPAQ REAR PANEL INTERFACE PCB 690D1586 (EFN01103)
+5V
ENCODER COM
LIMIT SUPPLY
LIMIT COM
100 LINE RIBBON 630C1962 (NPAQC-L)
100 LINE RIBBON 630C1963 (NPAQC-S)
A2 NPAQ CONTROL PCB 690D1585 (EFN01105) ULTRA
J11 CONNECTION TO A9 J4-2 AS PER ECN7346
A9 NPAQ MXR-6 EFN01123 690D1592
5 POS CONN ECK01075
A1 NPAQ DRIVE INTERFACE PCB 690D1585 (EFN01104)
IN REFERENCE TO ECN11740 VERIFY THAT TIE-WRAP IS PLACED AROUND TERMINALS OF BR2
J15 ADD KEY (EIK00501) TO PIN 1
J12 ADD KEY (EIK00501) TO PIN 2
BRAKE SUPPLY+ (+24V)
BRAKE SUPPLY- (GND)
A1-J13-1,3,4,5,6,8 FERRULE EIK01072
A1-J13-2,7 FERRULE EIK01070
A10 VOLTAGE SELECT BD. 690B1604 EFN01149
A11 HCS-6/DVP-A ECZ01010 +5V @ 6A
AC WIRING FROM A10-J1 TO BE ROUTED ALONGSIDE POWER SUPPLY BRACKET, SECURE WITH CABLE TIES (EIZ00100). WIRES MUST NOT PROTRUDE ABOVE BRACKET.
NPAQ MOTOR CABLE ASSY 630D1977 (NPAQC-MOTOR)
AXIS 1
AXIS 2
AXIS 3
AXIS 4
AXIS 5
AXIS 6
#16 BLK
#16 RED
#16 WHT
#16 GRN/Y

INSTALL LATEST VERSION 5 (CURRENT) FIRMWARE

CUSTOMER NAME: FERMILAB
ORDER NUMBER: 185744
APPROVED BY: AGS   RELEASE DATE: 10-13-2016
CUST FILE NAME: 185744 620D1332R1 NPAQ WIRING

REVISIONS				
REV.	ZONE	DESCRIPTION	DATE	DRAWN
A		ADDED ECN6583	12/02	JRF
B		ADDED ECN6770,6919,6937,7005, 7139,7220,7303,7375A,7487	01/04	JRF
C		ADDED ECN6755,7916,8096,8200,8280,8408	1/05	WG
D		ADDED ECN8732,8773,8808,8813B,8990,9016,9192	3/06	SMN
E		ADDED ECN9715	11/06	JRF
F		ADDED ECN9812,9881,10086A,10215,10238	7/07	JRF
G		ADDED ECN10332,10474,10480	10/07	JRF
H		ADDED ECN10565	10/07	WG
J		ADDED ECN10588,11421,11942	9/09	JRF/WG
K		ADDED ECN12060,12220,12589,12626	4/10	JRF
L		ADDED ECN12705,12766,12860	12/10	WG
M		ADDED ECN13141,13189,13913	3/12	JRF
N		ADDED ECN14951A,14994,15143A	3/14	AMO
P		ADDED ECN15894 (NO CHG THIS SHT)	8/14	AMO
Q		ADDED ECN16310	12/14	AMO
R		ADDED ECN16877	9/15	AMO

NOTES:

1. SHEET LIST: SHEET 1 & 2 ARE THE SCHEMATIC.
SHEET 3 IS THE ASSEMBLY.
SHEET 4 IS THE DEFINITION OF LAYERS FOR THE SCHEMATIC (SHT 1 & 2)
SHEET 7 IS THE EMI SHIELDING SHEET.

SYSTEM NUMBER 1 SERIAL LABEL

Aerotech.com
SYSTEM SERIAL NO: 185744-A-1-1
MODEL: NPAQ-A-80B-80B/ULTRA/
1-DP32030E/2-DP32020E/3-DP32020E/
4-DP32020E/5-DP32020E/6-DP32020E/
NO SPLIT/STD COOL/STAND/BRAKE-3/
ESTOP2/MXR-6/M6
DWG: 620D1332 REV R

2. PLACE SYSTEM SERIAL LABEL (EIA01252) ON REAR OF CHASSIS AS SHOWN ON ASSY DRAWING.
PLACE AC POWER INPUT LABEL SHOWN BELOW ON REAR OF CHASSIS AS SHOWN ON ASSEMBLY DRAWING.

TÜV C US 115 VAC 10 A 50/60Hz

EIA01444 (1/2 SCALE)

1) DO NOT PLACE LABEL WITH TUV SYMBOL ON NONSTANDARD NPAQ'S (ES VERSIONS) WITHOUT OBTAINING ENGINEERING APPROVAL.
2) IF LABEL IS MADE IN HOUSE CUT OUT LABEL TO SIZE SHOWN ON LABEL DRAWING BEFORE PLACING ON UNIT AS SHOWN ON ASSEMBLY.

3. VOLTAGE SELECTOR SWITCH CONFIGURATION.

	A6 SWITCHES				A10 SWITCHES	
	A6-S1	A6-S2	A6-S3	A6-S4	A10-S5	A10-S6
A-115VAC	UP	UP	UP	UP	UP	UP
B-230VAC	DN	UP	DN	UP	DN	UP
C-100VAC	UP	DN	UP	DN	UP	DN
D-200VAC	DN	DN	DN	DN	DN	DN

4. NPAQ DRIVE INTERFACE PCB (A1)

AXIS	1	2	3	4	5	6
AMPLIFIER	DP32030E	DP32020E	DP32020E	DP32020E	DP32020E	DP32020E

BUS 1 (80B) CONFIGURATION CHART:

T1	C1	C2	BW1
(EAX01006)	16000uF (ECC01002)	16000uF (ECC01002)	BW1 OUT
NONE	16000uF (ECC01002)	16000uF (ECC01002)	BW1 OUT

BUS 2 (80B) CONFIGURATION CHART:

T2	C3	C4	BW2
(EAX01006)	16000uF (ECC01002)	16000uF (ECC01002)	BW2 OUT
NONE	16000uF (ECC01002)	16000uF (ECC01002)	BW2 OUT

BUS SEPARATION:
NO SPLIT (STD):
IN: BW10, BW11, BW12, BW20, BW21, BW22,
IN: BW30, BW31, BW32, BW40, BW41, BW42,
IN: BW50, BW51, BW52

AUTOMATIC BRAKE "DE-ENERGIZED": JP1
A) +3.3V MONITORED: SET JP1 1-2 IN (STD)
BRAKE OPTION: INSERT M1 ODC5A (ECS000394).

5. NPAQ DSP CONTROL BD. (A2)
VERIFY JP1 IS SET TO 1-2*
M28 - HCPL-2601 (ECS00119) IS TO BE INSTALLED AT ALL TIMES.

RS232 / RS422 SELECT (SEE ALSO NOTE 7)	
RS232	JP2 1-2*
RS422	JP2 2-3

RESET TIMEOUT (JP3 MUST BE SET1-2)	
RESET TIMEOUT ENABLED	JP3 1-2*
RESET TIMEOUT DISABLED (FOR TEST ONLY)	JP3 2-3

AUX FIREWIRE BUS POWER SOURCE (NORMALLY NOT USED)	
EXTERNAL POWER SOURCE	JP4 1-2, JP5 1-2
+12 VOLT POWER SOURCE (A4) (DEFAULT)	JP4 2-3*, JP5 2-3*

S1 SWITCH SETTINGS		
S1-1 (RESERVED)	S1-1 OPEN (OFF)	S1-1 CLOSED (ON)
S1-2 (RESERVED)	S1-2 OPEN (OFF)	S1-2 CLOSED (ON)
S1-3 (RESERVED)	S1-3 OPEN (OFF)	S1-3 CLOSED (ON)
S1-4 (RESERVED)	S1-4 OPEN (OFF)	S1-4 CLOSED (ON)
S1-5 (RESERVED)	S1-5 OPEN (OFF)	S1-5 CLOSED (ON)
S1-6 (RESERVED)	S1-6 OPEN (OFF)	S1-6 CLOSED (ON)
S1-7 (RESERVED)	S1-7 OPEN (OFF)	S1-7 CLOSED (ON)
S1-8 ULTRA/BASE	S1-8 OPEN (OFF)	S1-8 CLOSED (ON)

S1-1 SWITCH IS NEAREST TO THE "S1" MARKING ON PCB.

21. PLACE WARNING LABEL (EIA01135) ON TOP COVER OF CHASSIS. SEE ASSEMBLY DWG (SHT 3) FOR LOCATION.

WARNING: DISCONNECT POWER BEFORE SERVICING
WARNING: HAZARDOUS VOLTAGES PRESENT UP TO 8 MINUTES AFTER POWER IS DISCONNECTED
CE ⚠ ⚠

EIA01135 (1/2 SCALE)

14. POWER SUPPLY (A11):
CUT OUT ANY WIRE JUMPERS BETWEEN TRANSFORMER PINS 1 THRU 5 (PINS MUST BE ISOLATED). SOLDER WIRES FROM A10 TO TRANSFORMER PINS 1 THRU 5.

8. USE 16-14 AWG FASTON CONNECTORS EIK00432. LABEL WITH PERMANENT MARKER.

9. SPLICE AND INSULATE WIRING TO/FROM TRANSFORMERS AS REQUIRED TO OBTAIN PROPER LENGTH.

10. ALL GRN/YEL GND WIRES TO BE CONNECTED TO SEPARATE CHASSIS MOUNTING POINTS. USE EIK00432. ADD PROTECTIVE GROUND LABEL (EIA00435) AT CONNECTION POINT IN CHASSIS.

11. ALL CRIMPED CONNECTIONS MUST ALSO BE SECURED WITH WIRE TIES OR HEATSHRINK. SOLDER CONNECTIONS SHOULD BE 'J' HOOKED.

12. ALL GRN/YEL GND WIRES TO BE CONNECTED TO SEPARATE CHASSIS MOUNTING POINTS. USE EIK00244. ADD PROTECTIVE GROUND LABEL (EIA00435) AT CONNECTION POINT IN CHASSIS.

13 ALL WIRING MUST NOT TOUCH BARE TERMINALS OR EXPOSED CIRCUITRY.

7. NPAQ REAR PANEL INTERFACE PCB (A3)

LIMIT POWER SUPPLY						
AXIS	1	2	3	4	5	6
	JP1	JP2	JP3	JP4	JP5	JP6
+5V (TRACE)	1-2*	1-2*	1-2*	1-2*	1-2*	1-2*
J16 INTERNAL	2-3	2-3	2-3	2-3	2-3	2-3

+5 V MIN. LOAD FOR A4 SUPPLY	
MXR NOT INSTALLED	JP7 IN *
MXR INSTALLED	JP7 OUT

AIN 2- SET FOR SINGLE (JOYSTICK) OR DIFFERENTIAL INPUT	
AIN 2- CONNECT TO GND (SINGLE, NEED FOR JOYSTICK)	JP14 IN*
AIN 2- NOT CONNECTED TO GND (DIFFERENTIAL INPUT)	JP14 OUT

AIN 3- SET FOR SINGLE (JOYSTICK) OR DIFFERENTIAL INPUT	
AIN 3- CONNECT TO GND (SINGLE, NEED FOR JOYSTICK)	JP15 IN*
AIN 3- NOT CONNECTED TO GND (DIFFERENTIAL INPUT)	JP15 OUT

PSO OUTPUT POLARITY	
PSO OUTPUT POLARITY NORMALLY OPEN	JP16 1-2*
PSO OUTPUT POLARITY NORMALLY CLOSED	JP16 2-3

RS232 / RS422 SETTINGS (SEE ALSO NOTE 5)			
	RS232 CONNECTION TO PC	RS232 TX/RX REVERSED	RS422
JP8	1-2*	2-3	1-2
JP9	1-2*	2-3	1-2
JP10	1-2*	1-2	2-3
JP11	1-2*	1-2	2-3
JP12	1-2*	1-2	2-3
JP13	1-2*	1-2	2-3

6. NPAQ DSP CONTROL BD. (A2)

FIREWIRE DEVICE # (S2 0-4, ON (CLOSED) = 0, OFF (OPEN) = 1)					
NPAQ # (BASE +5)	S2-4	S2-3	S2-2	S2-1	S2-0
1 THRU 6	ON 0	ON 0	ON 0	ON 0	ON 0
7 THRU 12	ON 0	ON 0	OFF 1	OFF 1	ON 0
13 THRU 18	ON 0	OFF 1	OFF 1	ON 0	ON 0
19 THRU 24	OFF 1	ON 0	ON 0	OFF 1	ON 0
25 THRU 30	OFF 1	OFF 1	ON 0	ON 0	ON 0

S2-5 THRU S2-9 SWITCH SETTINGS (ON = CLOSED, OFF = OPEN)		
S2-5 (UNASSIGNED)	S2-5 OPEN	S2-5 CLOSED
S2-6 (UNASSIGNED)	S2-6 OPEN	S2-6 CLOSED
S2-7 (UNASSIGNED)	S2-7 OPEN	S2-7 CLOSED
S2-8 (UNASSIGNED)	S2-8 OPEN	S2-8 CLOSED
S2-9 (UNASSIGNED)	S2-9 OPEN	S2-9 CLOSED

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TOLERANCES ON

FRACTIONS		DECIMALS		ANGLES
0 TO 6	± 1/32	.XX	± .01	± 0-30'
OVER 6	± 1/16	.XXX	± .005	

RMS FINISH ✓ EXCEPT AS NOTED

NPAQ -A -80B -80B /ULTRA /1 DP32030E /2 DP32020E /3 DP32020E /4 DP32020E /5 DP32020E /6 DP32020E SPLITBUS 1-6 /STD COOL
LINECORD-US-115VAC /STD MOUNT BRAKE-3
MXR-6 ESTOP2
M6

DRAWN	JRF	11/02	PART CODE -	SIZE D	620D1332	REV. R
CHECKED	GLK	11/02				
APPD.	ARW	11/02	SCALE NTS	FILE NAME 620D1332R1.dwg	SHEET 1 OF 7	